Final Lecture: Infinite Relational Model

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1 The Infinite Relational Model

What happens when you want to cluster your data, but the number of clusters is unknown? While some approaches involve fitting several models with a varying number of clusters to your data and comparing model fit statistics, the Bayesian approach is to specify a model which is allowed to dynamically grow the number of clusters as the complexity of the data warrants.

The **Infinite Relational Model** is the prototypical example of such a model.

> Kemp et al. (2006). Learning Systems of Concepts with an Infinite Relational Model[1]

For this last project, I coded up a simple version of Charles Kemp's Infinite Relational Model (IRM) in `irm.R` to co-cluster rows and columns of a simple 2-dimensional binary relation.

1.1 Demo

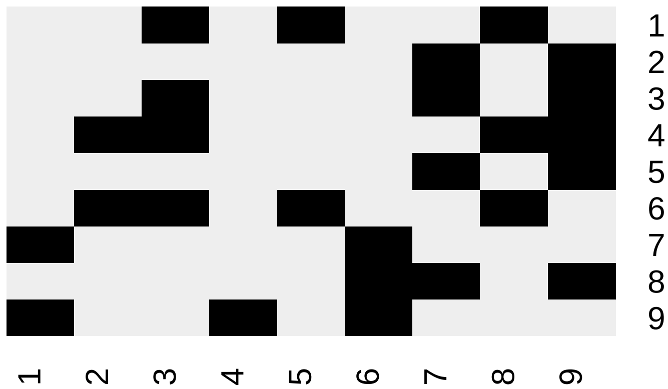
```
source('irm.R')
```

1.1.1 Sanity check

As a sanity check, we use the toy matrix in the original paper:

```
R = rbind(
  c(0, 0, 1, 0, 1, 0, 0, 1, 0),
  c(0, 0, 0, 0, 0, 0, 1, 0, 1),
  c(0, 0, 1, 0, 0, 0, 1, 0, 1),
  c(0, 1, 1, 0, 0, 0, 0, 1, 1),
  c(0, 0, 0, 0, 0, 0, 1, 0, 1),
  c(0, 1, 1, 0, 1, 0, 0, 1, 0),
  c(1, 0, 0, 0, 0, 1, 0, 0, 0),
  c(0, 0, 0, 0, 0, 1, 1, 0, 1),
  c(1, 0, 0, 1, 0, 1, 0, 0, 0)
)
plot.R(R)
```

[1] http://web.mit.edu/cocosci/Papers/Kemp-etal-AAAI06.pdf




```
Z = irm(R, sweeps = 1000)
top.n(Z)
#> 122121323 122123424 122121324 122321424 122324525 122323424 123121424 122121343
#>       895        35        23        10         6         5         4         3
#> 122321425 112121323
#>         3         2
plot.R(R, mode.irm(Z))
```

This successfully finds the clusters of rows and columns that correspond to the original paper.